Lecture Content-3 (Week-3)

Course Code with Title: MIS0611401: Information Systems Management (**B.Sc in CSE**)

Topic: Information Systems, Organizations, and Strategy

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Information Systems, Organizations, and Strategy

This chapter / Lecture content explores the **relationship between information systems (IS), organizational structures, and business strategy**. It emphasizes that IS is not just a technical tool but a strategic enabler that reshapes how firms compete, organize, and innovate.

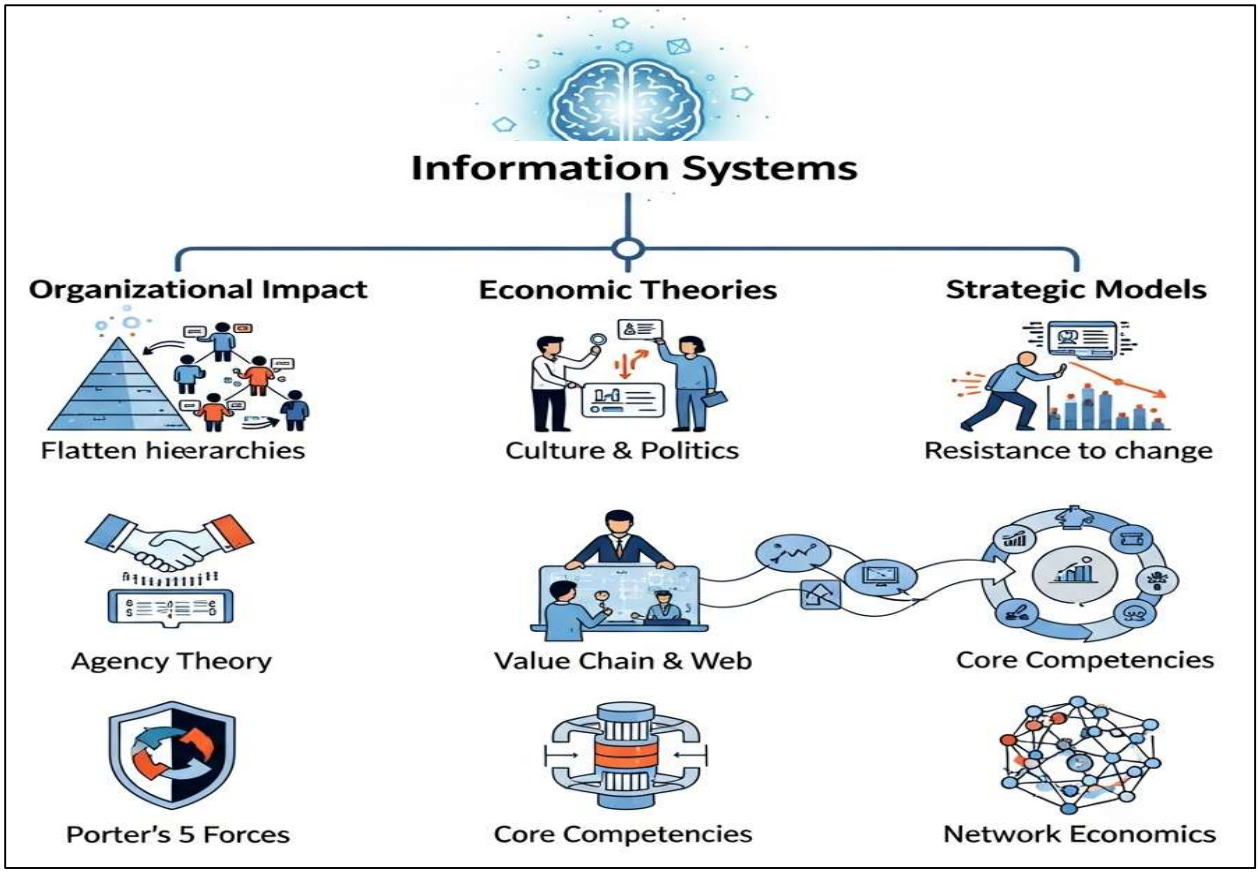


Figure: IS, Organizations, and Strategy

This figure shows IS at the center, influencing organizations through structure and culture, reshaping economics by lowering costs, and driving strategy through competitive models. Together, these dimensions explain why IS is both a **technical enabler** and a **strategic weapon** in modern firms.

1. Organizations and Information Systems

Organizations as Social Structures
Organizations transform **inputs** (capital, labor, information) into **outputs** (goods and services). They rely on routines, hierarchies, and culture to maintain stability. IS becomes embedded in these structures, influencing how work is performed and how decisions are made.

Technical View of IS
From a technical perspective, IS is seen as an **input-output machine**:

- It collects data, processes it, and produces information.
- It supports efficiency by automating tasks and standardizing workflows.
- Example: **Unilever** uses IS to coordinate global supply chains, ensuring that production, marketing, and finance are aligned across multiple countries.

Behavioral View of IS
From a behavioral perspective, IS reflects the **people, culture, and politics** within organizations:

- IS adoption depends on organizational culture and willingness to change.
- Resistance may occur if employees feel threatened by automation or transparency.
- Example: **Grameenphone** in Bangladesh leverages IS to manage nationwide telecom operations, but success depends on employee collaboration and customer trust.

Impacts of IS on Organizations
IS reshapes organizational structures and processes in several ways:

- **Flattening hierarchies:** Real-time information reduces the need for middle managers, empowering employees at lower levels.
- **Reducing transaction costs:** IS lowers costs of searching, contracting, and coordinating, enabling leaner organizations.
- **Empowering employees:** Dashboards and analytics provide real-time insights, allowing faster and more confident decision-making.
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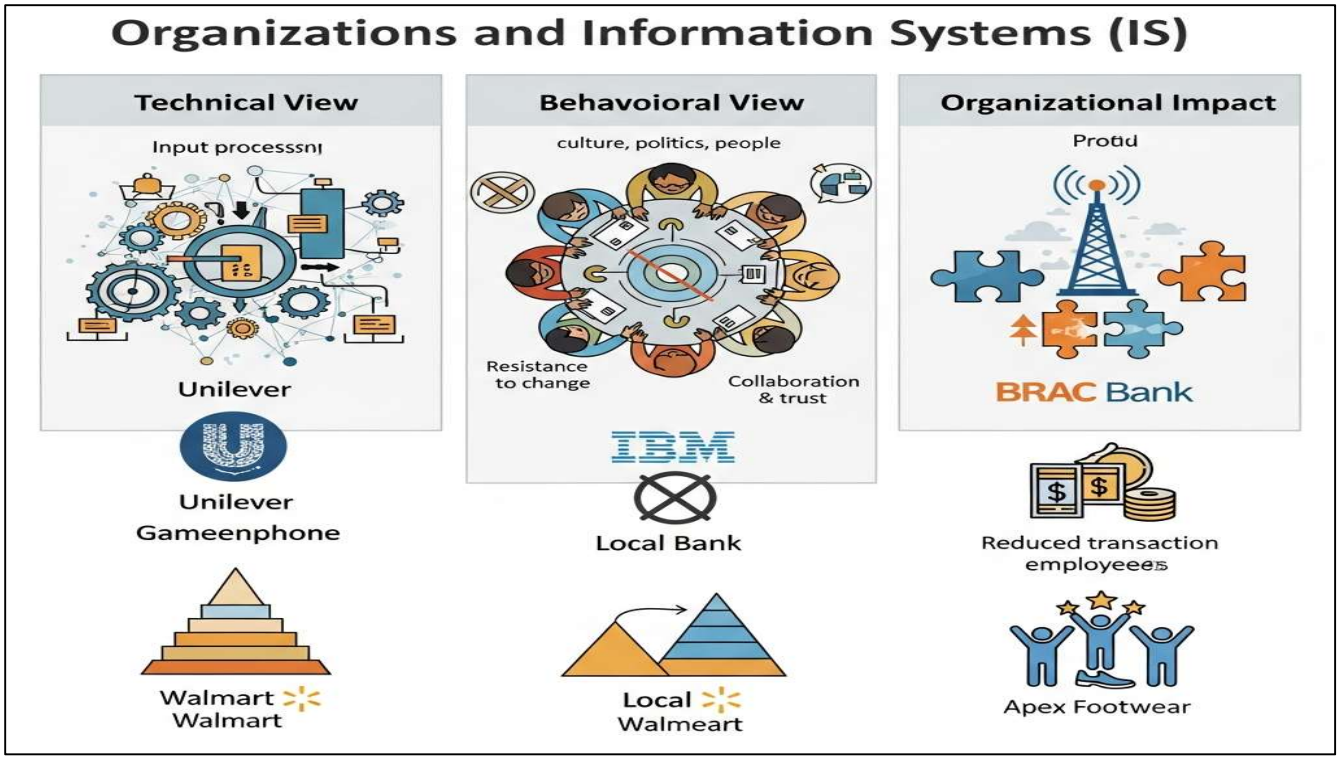


Figure: Organizations and IS

This Figure shows IS at the center of organizational life. The **technical view** emphasizes efficiency and automation, while the **behavioral view** highlights culture and human dynamics. The **organizational impact** demonstrates how IS flattens hierarchies, reduces costs, and empowers employees. Together, these perspectives explain why IS is both a **technical enabler** and a **social system** that must align with organizational values and strategies.

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2. Economic Theories (Information Systems and Organizations)

This section explains how **economic theories** help us understand the impact of information systems (IS) on organizations. IS does not only automate tasks—it fundamentally changes the economics of how firms operate, reducing costs and enabling new organizational structures.

Transaction Cost Theory

- **Definition:** Transaction costs are the expenses of conducting business, such as searching for suppliers, negotiating contracts, and coordinating logistics.
- **Impact of IS:** IS reduces these costs by automating processes, providing real-time data, and integrating workflows.
- **Global Example: Walmart** uses IS to minimize transaction costs in its supply chain. Its real-time inventory management system ensures products are restocked efficiently, reducing the need for costly manual coordination.
- **Bangladeshi Example: Apex Footwear** integrates IS into its production and distribution processes. By digitizing workflows, Apex reduces delays, lowers coordination costs, and ensures timely delivery to retailers.

Agency Theory

- **Definition:** Agency theory deals with the relationship between principals (owners) and agents (managers). Monitoring agents is costly because of information asymmetry.
- **Impact of IS:** IS lowers monitoring costs by providing dashboards, analytics, and reporting tools. Managers can oversee larger organizations with fewer resources.
- **Global Example: Amazon** uses IS dashboards to monitor global operations, ensuring managers can track performance across multiple regions without excessive overhead.
- **Bangladeshi Example: Grameenphone** leverages IS to monitor telecom operations nationwide. Managers can track customer service metrics and network performance in real time, reducing the need for layers of supervision.

Organizational Implications

- **Lean Structures:** IS enables firms to operate with fewer middle managers by reducing transaction and monitoring costs.
- **Outsourcing:** Lower transaction costs make outsourcing viable, as coordination with external partners becomes easier.
- **Empowered Employees:** Real-time information allows employees at lower levels to make decisions confidently, flattening hierarchies.

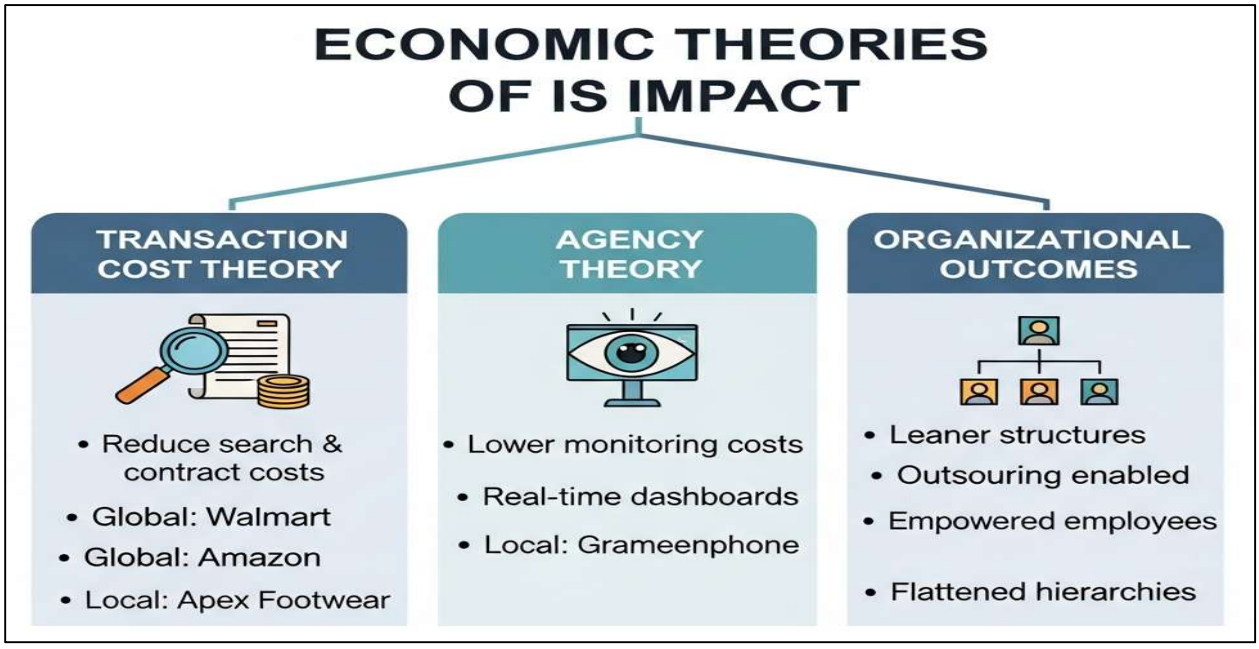


Figure: Economic Theories and Information Systems

This figure shows how IS reduces both **transaction costs** and **agency costs**, leading to leaner, more efficient organizations. Global firms like Walmart and Amazon demonstrate large-scale applications, while Bangladeshi firms like Apex Footwear and Grameenphone illustrate localized implementations. The result is **greater efficiency, scalability, and empowerment**, making IS a cornerstone of modern organizational strategy.

3. Strategic Role of Information Systems

Information Systems (IS) are not just operational tools; they are **strategic weapons** that help firms gain competitive advantage. By reshaping how organizations interact with competitors, customers, suppliers, and substitutes, IS enables firms to innovate, differentiate, and scale globally.

Porter's Competitive Forces Model

IS empowers firms to respond to the five forces shaping competition:

1. **Competitors:** IS enables differentiation through digital services. For example, **Netflix** uses recommendation algorithms to stand out from competitors. In Bangladesh, **Chaldal** leverages IS to provide personalized grocery delivery services.
2. **New Entrants:** Online platforms lower barriers to entry. IS allows startups to compete with established firms by offering digital-first services.
3. **Substitutes:** Innovation through IS reduces the risk of replacement. For instance, **Spotify** disrupted traditional music distribution with streaming.
4. **Customers:** CRM systems strengthen loyalty by personalizing experiences. **Amazon** uses IS to track customer preferences, while **bKash** in Bangladesh tailors mobile financial services to customer needs.
5. **Suppliers:** SCM systems integrate supply chains, improving efficiency and reducing costs. **Walmart** uses IS to coordinate suppliers globally, while **Apex Footwear** in Bangladesh integrates IS to manage raw materials and distribution.

Value Chain Model

IS adds value across both primary and support activities:

- **Primary Activities:** Logistics, operations, outbound logistics, marketing, and service. IS ensures efficiency and customer satisfaction.
- **Support Activities:** HR, procurement, and technology development. IS integrates these functions to support overall business goals.
  Example: **Unilever** uses IS to streamline procurement and HR globally, while **Square Pharmaceuticals** in Bangladesh integrates IS into production and compliance processes.

Value Web

IS creates **flexible networks of firms** that collaborate and innovate together.

- Globally, **Apple's ecosystem** integrates suppliers, developers, and customers.
- In Bangladesh, **Pathao** operates as a value web, connecting drivers, riders, restaurants, and customers through a digital platform.

Core Competencies & Synergies

IS strengthens unique skills and integrates business units:

- **Core Competencies:** IS helps firms focus on what they do best, such as innovation or customer service.
- **Synergies:** IS integrates different business units, creating efficiencies and shared knowledge.
  Example: **Google** leverages IS to integrate search, advertising, and cloud services, while

Grameenphone integrates telecom services with mobile financial platforms in Bangladesh.

Network Economics
IS increases value as more users join a network.

- **Global Example:** Social platforms like **Facebook** and marketplaces like **Amazon** gain more value as user bases expand.
- **Bangladeshi Example: bKash** benefits from network economics—the more users adopt mobile payments, the more valuable the platform becomes for customers, merchants, and partners.

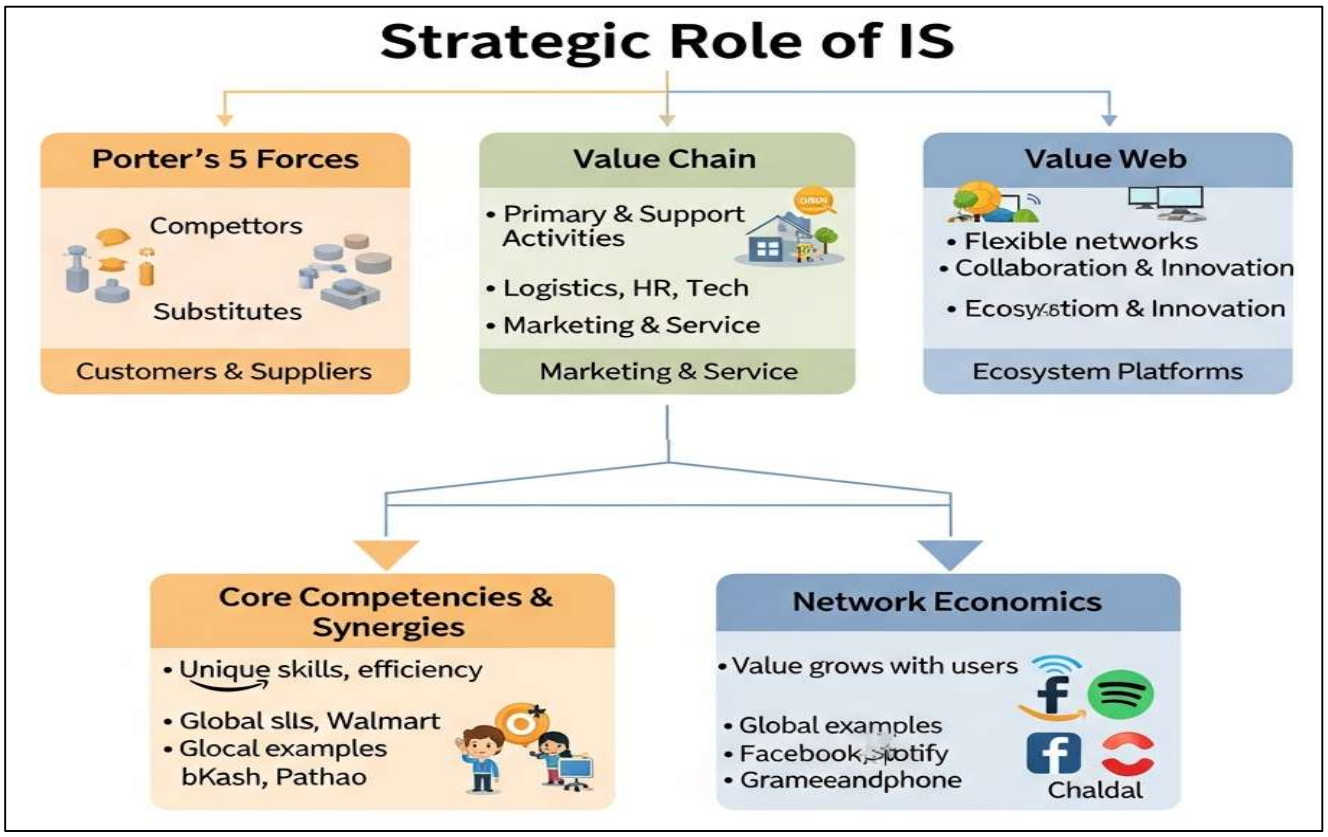


Figure: Strategic role of Information systems

This figure shows IS at the center of strategy. Through **Porter's model**, IS helps firms respond to competitive forces. The **value chain** highlights how IS adds efficiency across activities, while the **value web** demonstrates collaboration across ecosystems. IS also strengthens **core competencies**, creates **synergies**, and leverages **network economics** to scale value as user bases grow.

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4. Business Ecosystems and Platforms

This section highlights how **Information Systems (IS)** enable the creation of **business ecosystems and digital platforms**, where multiple firms collaborate, share resources, and innovate together.

Unlike traditional linear supply chains, ecosystems are dynamic networks of interconnected organizations, coordinated through IS.

Definition and Role

- **Business Ecosystem:** A network of suppliers, distributors, customers, competitors, and government agencies linked through IS.
- **Digital Platforms:** Technological foundations (like Amazon Marketplace or Uber) that allow participants to interact, transact, and innovate.
- **Strategic Importance:** IS makes ecosystems flexible, scalable, and responsive, allowing firms to adapt quickly to market changes.

Global Examples

- **Amazon Marketplace:** Connects millions of sellers and buyers worldwide. IS integrates logistics, payments, and customer service, creating a seamless ecosystem.
- **Uber:** Uses IS to coordinate drivers, riders, and payment systems in real time. Its platform is a digital ecosystem that scales globally.
- **Apple Ecosystem:** Integrates hardware, software, and services, creating value through interconnected devices and applications.

Bangladeshi Examples

- **Pathao:** Operates as a digital ecosystem connecting drivers, riders, restaurants, and customers. IS enables real-time ride-hailing, food delivery, and payment integration.
- **bKash:** Functions as a financial ecosystem, linking customers, merchants, and banks. IS ensures secure mobile transactions, enabling financial inclusion across Bangladesh.
- **Daraz:** As an e-commerce platform, Daraz connects sellers and buyers, integrating logistics and digital payments through IS.

Benefits of Ecosystems and Platforms

- **Flexibility:** Firms can adapt quickly to market changes.
- **Innovation:** Ecosystems foster collaboration and new ideas.
- **Scalability:** Platforms grow rapidly as more participants join.
- **Efficiency:** IS reduces transaction costs and improves coordination.
- **Customer Value:** Ecosystems provide integrated services, enhancing user experience.

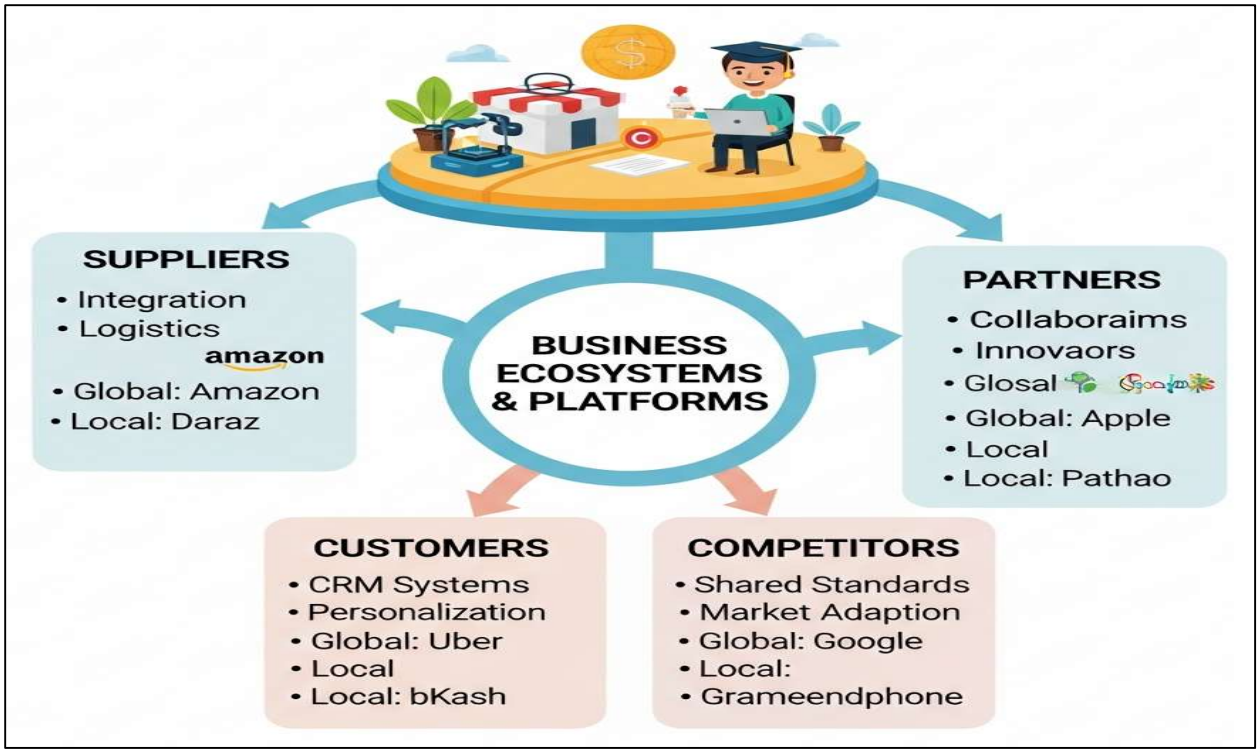


Figure: Business Ecosystems and platforms

This figure shows IS at the center of ecosystems, connecting suppliers, customers, partners, and even competitors. Global firms like Amazon, Uber, and Apple demonstrate large-scale ecosystems, while Bangladeshi firms like Pathao, bKash, and Daraz illustrate localized applications. The arrows represent **flows of information and transactions**, highlighting how IS enables collaboration, innovation, and scalability.

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5. Challenges of Strategic Information Systems

Strategic Information Systems (IS) are powerful tools, but their implementation comes with significant challenges. These challenges revolve around **alignment with business goals**, **resistance from organizational culture**, and **ethics and security concerns**.

Alignment

- **Definition:** IS must align with the overall business strategy to deliver value. If systems are implemented without clear alignment, they risk becoming costly investments with little impact.
- **Global Example: Walmart** aligns IS with its low-cost leadership strategy by using real-time inventory systems to reduce costs and improve efficiency.
- **Bangladeshi Example: bKash** aligns IS with its mission of financial inclusion by integrating mobile payment systems that serve millions of unbanked citizens.
- **Key Insight:** Alignment ensures that IS investments directly support organizational objectives, whether cost leadership, differentiation, or innovation.

Resistance

- **Definition:** Organizational culture may resist change, especially when IS disrupts established routines or threatens job roles.
- **Global Example: General Motors** faced resistance when introducing ERP systems, as employees were reluctant to abandon legacy processes.
- **Bangladeshi Example: Grameenphone** encountered cultural resistance when shifting to digital collaboration platforms, requiring training and change management to ensure adoption.
- **Key Insight:** Resistance can be mitigated through leadership, communication, and employee involvement in system design and rollout.

Ethics & Security

- **Definition:** Strategic IS raises issues of privacy, fairness, and data protection. Firms must safeguard customer data and ensure ethical use of information.
- **Global Example: Facebook** has faced scrutiny over data privacy and ethical use of customer information.
- **Bangladeshi Example: BRAC Bank** must ensure compliance with data protection laws when using IS for customer relationship management.
- **Key Insight:** Strong governance, encryption, and ethical policies are essential to maintain trust and avoid reputational damage.



Figure: Challenges of Strategic Information systems

This figure shows the three pillars of challenges: **alignment**, **resistance**, and **ethics/security**. Global firms like Walmart, GM, and Facebook illustrate large-scale challenges, while Bangladeshi firms like bKash, Grameenphone, and BRAC Bank demonstrate localized issues. The key takeaway is that strategic IS must be carefully managed to ensure alignment with goals, overcome cultural resistance, and uphold ethical standards.

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